Lesson Plan · Thursday, October 29

Algebra I · Mr. Pandey · Module 2, Topic 2, Lesson 3: Determining Slopes of Perpendicular Lines (Day 2) · essentials only

Objectives (students will…)

- Write the equation of a line perpendicular to a given line through a given point.
- Find the slope from two points first, then the perpendicular slope.
- Show what they know on the Form B exit ticket (like the End of Topic Assessment).

TEKS

- **A.2F** perpendicular line through a point **A.2E** parallel line through a point **A.2G** horizontal / vertical lines
- **A.3E** transformations of f(x) = x (exit ticket Q1, 4, 5, 6)

Materials

- ☐ Bell work copies: *BELL WORK/T2-Bell-Work-Thu-10-29.pdf* (same questions as *t2-lesson3-bellwork-day2*)
- ☐ Slides: *t2-lesson3-perpendicular-slopes-catchup* (Day 2 slides) + "Lesson 3 with answers"
- ☐ Student Edition pp. 312–313
- ☐ IEP/504 students' unfinished Wednesday tickets
- ☐ Exit ticket copies: *EXIT TICKET/T2-Lesson3-Exit-Ticket.pdf* (Form B, 2-sided) + answer key

Agenda (60 minutes)

Time	Activity	What to do / say
0–5	**Bell work** (4 questions)	Perpendicular slope of ⅔, slope of a horizontal line, vertical line through (2, −1), perpendicular line through (4, 1). Go over **Q4**: it is today's lesson. IEP/504 students may finish Wednesday's exit ticket instead.
5–30	**Activity 3.3** ★ Writing Equations of Perpendicular Lines · SE pp. 312–313	Model **Question 1**: ⊥ to y = 2x + 1 through (6, 2) → slope −½ → y = −½x + 5. Pairs do **Question 2** (slope is −¾, so use 4/3 → y = 4/3x − 12) and **Question 5** (meet at (5, 9), other point (−4, 4) → y = −9/5x + 18). Pairs check Wednesday's Q3–4 (y = ⅓x and y = −2x + 3). Steps on the board: 1. slope of given line · 2. flip + change sign · 3. point-slope · 4. solve for y · 5. check the point
30–55	**Exit ticket, Form B** (10 questions, independent)	Same 10 items and order as the End of Topic Assessment, new numbers (Wednesday was Form A). Front and back, 25 minutes. IEP/504: read aloud on request, reminders card, extra time.
55–60	**Wrap up**	Collect tickets. Homework: **Lesson 3 homework** and **Lesson 3 quiz** on the student site. Next class Mon 11/2: review day (practice test); test Tue 11/3.

Check for understanding

- Whiteboards during 3.3: "Given slope −3. Perpendicular slope?" (⅓)
- Thumbs: "y = 4x and y = −4x are perpendicular." (Down: 4 · −4 = −16.)
- Compare Form A (Wed) and Form B (Thu) scores question by question.

After class (using the answer key)

- Missed Q1, 4, 5, 6 → Lesson 1 or 2 video.
- Missed Q7, 8 → Lesson 3 video + homework tonight.
- Missed Q2, 3, 9, 10 → small group on Mon 11/2 review day.

Skipped (not essential): Talk the Talk (SE p. 314) and the Topic 1 Lesson 6 catch-up. TE wording slip: Activity 3.3 Question 2's given slope is −¾ (the teacher guide says ¾).